Disruptive effect of an antimicrobial peptide from *Acropora digitifera* on the structure of the AcrAB-TolC efflux pump of *Escherichia coli*.

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INTRODUCTION

In 2019, bacterial antibiotic resistance was caused 1.27 million deaths globally, with 50% attributed to Gram-negative bacteria (Murray et al., 2022). Efflux proteins play a key role in both intrinsic and acquired resistance in bacteria (Du et al., 2018).

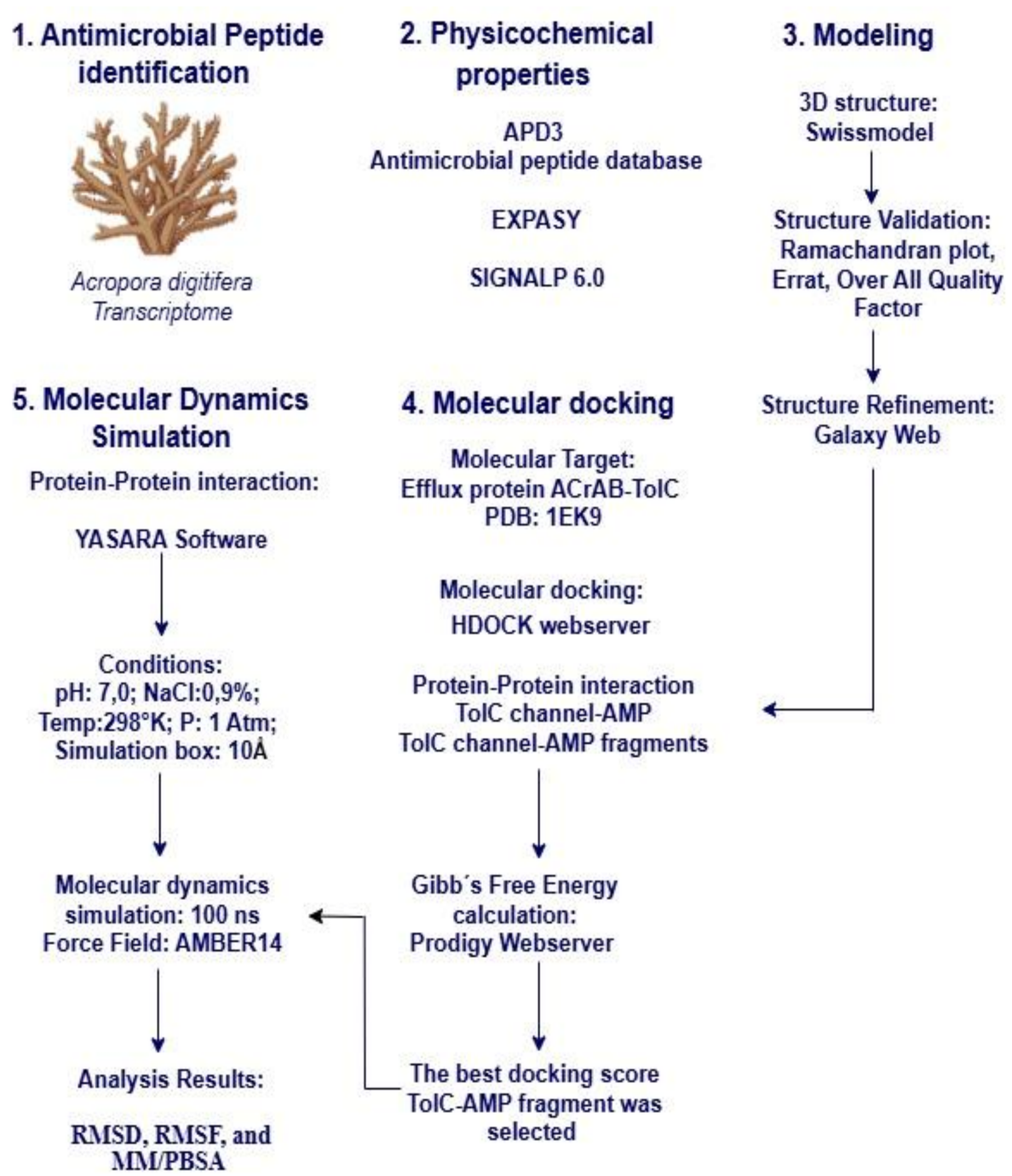
In Gram-negative bacteria, Resistance-Nodulation-Division (RND) family efflux proteins form a tripartite structure that transports molecules like antibiotics, dyes, and detergents, out of the cell, contributing to multidrug resistance, virulence, and biofilm formation (Blair & Piddock, 2009; Athar et al., 2023).

Due to their role in multidrug resistance, RND efflux proteins are targets for antibacterial treatment development (Blair & Piddock, 2009).

Antimicrobial peptides (AMPs), particularly those derived from marine organisms show promise as broad-spectrum, stable, low-toxicity antibaceterial agents.

This study focuses on antimicrobial peptides from the coral *Acropora digitifera* to asses their potential against the AcrAB-TolC efflux protein in *Escherichia coli*

METHODS



RESULTS

In a previous study, a 48 amino-acid antimicrobial peptide was identified in the transcriptome of the *Acropora digitifera* blastula stage (Uribe et al. In preparation). From the full-length peptide, five fragments were generated, and their 3D structures and physicochemical properties were characterized (Table 1).

The molecular docking results showed that the complete peptide (CP) exhibited a binding energy of -140.04. A 15-amino-acid fragment displayed the best binding affinity (Table 1).

Peptide length	Molecular Weight	theoretical isoelectric point	Total Net Charge	Percent hydrophobicity	Binding Energy	Ligand RMSD	ΔG Kcal/mol
48 aa	5258.03	8.01	+1	33%	-140,04	60,57	-14,5
32aa	3687,31	9,16	+2	36%	-175,02	64,53	-14,9
23 aa	2596,97	9,31	+2	295	-148,32	75,29	-12,3
19 aa	2101,38	5,79	0	35%	-165,12	70,84	-10
15 aa	1820,12	9,82	+3	31%	-178,18	74,33	-11,9
14 aa	1588,73	4,56	-1	27%	-173,64	73,84	-9,2

Table 1.Physicochemical properties and molecular docking results.

During the 100 ns simulation, the AcrAB-TolC efflux protein did not exhibit structural changes. At 90 ns, the CP-TolC complex showed an alteration in the TolC protein channel structure in the region of residues 259 to 272, suggesting a potential disruptive mechanism of action (Fig 1)

The 15 aminoacid fragment-TolC complex destabilized at 60 ns of simulation, generating a rupture in the channel at a beta sheet level in the region of residues 260 to 300 (Fig.2)

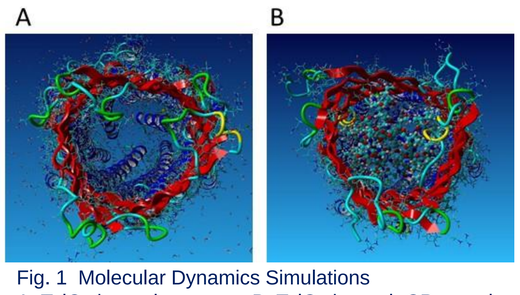


Fig. 1 Molecular Dynamics Simulations
A. TolC channel B. TolC channel- CP complex

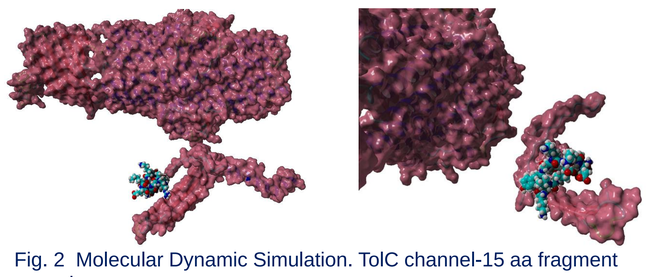


Fig. 2 Molecular Dynamic Simulation. TolC channel-15 aa fragment complex

DISCUSSION

The disruptive effect of the TolC-CP complex resembles that reported by Wang et al. (2022), in which the interaction between three selected AMPs from the APD3 database and the TolC channel of *Candidatus Liberibacter asiaticus* was found to reduce the surface area at the efflux pump outlet, potentially inhibiting its function.

The RND-type complex depends on each protein component for effective drug efflux; the absence of any element disrupts both structure and function(Athar et al.,2023).This disruption in efflux protein structure, resulting from the interaction between the TolC channel and the 15-amino-acid fragment, directly interferes with its functionality.

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ACKNOWLEDGMENTS

Universidad de la Amazonia
BIOS Centro de Bioinformática y Biología Computacional
Universidad Tecnológica de Pereira
Universidad de Manizales